

RSM — The Structural Chain (v9.2)

Logic only. Status tags mark epistemic standing: [derived] follows from the premises; [candidate] proposed, not forced; [open] named and unresolved; [rendering] a fact about a presentation, kept as an indexed reading and not load-bearing. Not sealed.

Cross-references name their item and give the number in parentheses; the name is authoritative.

Premises

P1 — the Conditional. Reality is infinitely vast and infinitely divisible: no terminal resolution in either direction — no edge outward, no floor inward.

P2 — the Bridge. What obtains is distinguishable. Contrapositively: what is not distinguishable does not obtain.

Everything below is what these two premises jointly require. Where a further assumption operates, it is named in the open items.

Definition (frame). A frame R_n is the coinherence of one distinction: two conjugate modes, their reciprocal relation, their balance, and their origin — arriving together or not at all. One constitutive pair per frame.

Corollary. Every distinction constitutes a frame; distinction beneath a frame's unit is therefore another frame, not a smaller item within this one.

1. The total paradox: P_0

P_0 is the total paradox: the condition in which distinction and indistinction — the master conjugate pair, each constituted by the other — stand equal and cancel. It is not indistinction: indistinction is one side of the pair, and P_0 privileges neither side; it is the pair's collapse. Its presenting faces are **|0|** (the collapse via emptiness — nothing to distinguish) and **|1|** (the collapse via fullness — nothing standing out; the bars guard the reading: never the numeral, never any frame's unit 1_n). One pair at equality within a frame is a P_n — met at resolution, because the frame's other distinctions carry the load. P_0 is the case with no frame and no load-bearer: every pair at equality at once, including the pair that every pair presupposes. By P2 it cannot obtain — at the total cancellation nothing is distinguishable — and it cannot even be asserted without contradiction, since stating the equality

distinguishes its terms. **P₀ and P_n are one species — paradox, a conjugate pair at equality — at two scopes.** *[derived]*

Face-unity. The two faces are the two counterfactual terminals of one journey along the gradient: toward all-form-no-void and toward all-void-no-form. Neither is reached; the journey is one. Both are excluded in-frame, at the gradient's two ends, by the law's own shape (§4). *[derived]*

2. The ground

Because P₀ cannot obtain, at least one distinction obtains — an origin, a frame: O₁, R₁. The ground relation is the joint relation at zero scope: every origin is a paradox read from below, and O₁ reads P₀. Because P₀ is unframed, R₁ has no parent: each obtained tree is rooted. The ground's mechanism (identity through opposition — required directly by the impossibility) differs from every later frame's (generation at a joint); both are real, and the structure needs both. *[derived]*

At least one is what the impossibility gives. Nothing here counts the roots, and nothing here excludes more than one.

The two senses of smaller. Two distinct relations must not share a word:

- **Fineness** — motion along the gradient toward a mode's small end. In-frame, unbounded, terminal excluded by the law.
- **Descent** — the constitution of a child at a joint. Across-frame, inward only, not motion along anything.

Vastness and fineness are the gradient's two ends, both in-frame, exchanged by the mode-swap (math chain). Descent is neither of them. *[derived]*

3. The coinherence

- A distinction entails its two co-obtaining modes (X, Y) — one distinction, two modes, each constituted by the other; nothing prior was divided.
- **The modes are magnitudes. The conjugate of a mode is another mode, not its negative.** A negative would name something that both obtains and anti-obtains; what holds instead is that a mode is incoherent without its contrasting mode. A sign is therefore an **indexed comparison** — this, compared to what it is being compared to — and its implication varies with the reference chosen. The signed register is a legitimate indexed reading carrying no structural load. *[derived; §8]*
- A mode can be noticed without its conjugate having been measured; its **amount** exists

only as a ratio to its conjugate. Existence-dependence is not measurement-dependence.

- Both modes are measured from one reference — the point at which both would vanish. That is the origin O: a locus, never a position; nothing stands there.

4. The frame; the unit; the floor

1_n is the constitutive distinction's own magnitude — the conserved product of *the* pair that constitutes frame n — and simultaneously the frame's minimal unit: nothing beneath the constitutive magnitude obtains *within* frame n , because any distinction beneath it is, by the Corollary, another frame. "Sub- 1_n " is unit-relative typing, not size: magnitude comparison across frames is not defined, and the child is not smaller — the child is the approach that does not finish, continuing. The floor is the boundary of the frame's own definition, not an added restriction. *[derived given the reciprocity joint]*

The law. The two modes' amounts are locked: their product is conserved, $X \cdot Y = 1_n$. A conserved sum fails twice — it admits the terminals (one mode at nothing, the other at everything: P_0 's faces as attainable values) and it caps both modes, contradicting infinite reach. The product's curve has no endpoints and no ceilings; the exclusion of the impossible is the law's own shape, not a rule imposed on it. The native coordinates are fixed by the impossibility itself: they are the ones in which P_0 's faces sit at *nothing* and *without limit*. The constant is the unit, and the unit is the frame's reality: things differ in their proportion of the modes; reality is the invariant the proportions hold. *[derived; the joint's full derivation is realized in the math chain; its native-magnitudes step is open]*

Scale. Within a frame, scale is motion along the curve: rescaling one mode by λ and the other by $1/\lambda$, product invariant — the complete set of rescalings of one reality. No rescaling reaches the center. Scale is within-frame; the lineage is not scale. *[derived]*

5. The two centers

- **O = the co-vanishing locus** (unmet in its own frame's reading). O is where both modes would reach operational absence together; nothing on the curve approaches it — the curve's nearest locus to the center is the balance. The curve bends around O: the connection between the mode-dominant extremes is the curve itself, and "no through, only around" is a description of it, not an added rule. *[derived]*
- **P = the balance paradox.** By continuity the balance locus is entailed, and balance *at frame resolution* obtains: equal amounts of the two modes is not the modes becoming one. What cannot obtain is **exactness** — equality carried through every depth is a terminal resolution, and P1 denies terminals; beneath any frame's floor the equality

opens into finer-grained imbalance the frame cannot state. And P cannot serve as *frame n's origin*: one constitutive pair per frame, and n has no spare pair to grant it. Met at resolution; no exactness anywhere; no origin here. *[derived]*

- "Met"/"unmet" are frame-reading statuses only. Occupancy language does not type these loci.

6. The engine

The demand: continuity entails the balance locus, and balance-at-resolution obtains — the practical register is real. Denial A (origin): the locus cannot be frame n's origin. Denial B (exactness): exact balance cannot obtain. Must exist; met at resolution; no origin here; no exactness anywhere — simultaneously, locally. The demand is satisfied in the one mode of existence left to such a center: **existence-as-organizing-locus** — in-frame, as the locus circulation organizes around; trans-frame, as the origin a new structure organizes around. **The child frame is the pursuit of exactness, continuing below the floor:** the residue of imbalance beneath 1_n — the not-quite-balance the frame cannot state — is sub-unit distinction, hence a frame, whose origin is this very locus read from below. Recursion is exactness, permanently deferred. *[derived]*

Generation has no address. Constitution is not sited and not timed. The residue of imbalance is standing, so the child is entailed, not triggered — nothing occurs at perfect balance, because exactness never obtains and an inexactness is not a place. Parametric facts at balance-at-resolution are real and belong to the practical register; they are not the constitution. *[derived]*

Rendering guard. A drawing shows a smooth curve through an unproblematic point; the constitution-layer facts (no origin here; no exactness anywhere) have no rendering. The drawing's verdict never overwrites the constitution's.

7. Recursion

One locus, two frame-readings: a point in the parent's register, an origin in the child's — $O_{(n+1)}$ reads P_n . Nothing in the parent selects the child's orientation: a preference would be a distinction, and the parent has nothing to constitute it with. The joint, read as potential, is therefore a **ring** of orientations, no one privileged; what obtains is always one selection (all at once would be saturation), and different selections are different branches. The nesting is a tree, not a ladder — and a rooted tree: index-relationality holds within it (every frame reads itself as "n"; the labels are gauge), but no relabeling makes a root a child, because a root's parent-side is P_0 , an impossibility, not a frame. *[derived; branching at loci other than the balance: open]*

What a child shares with its parent. Not magnitude (comparison across frames is undefined), not orientation (nothing in the parent selects it), not content. What is shared is the frame-form — every frame is a frame — and, between this child and this parent specifically, the joint: **what the parent cannot state is what the child measures from.** The parent's inexactness is the child's reference. The relation is constitutive and standing, not a transfer and not an event. *[derived; whether any quantitative relation accompanies it is the cross-frame unit relation, open]*

8. The two tests

Every claim passes two tests before it enters.

The chart test: does it survive changing the drawing? Failed by: perpendicularity, straightness-as-driver, ambient space for the sphere, the sign on a mode — each a fact about a presentation.

The grammar test: does the word's grammar assert what the framework denies? Agentive verbs for structural facts, transfer-language for non-events, removal narratives for constitutive exclusions, staging conjunctions for simultaneous constitution, reified infinities, occupancy language for the loci, address-language for constitution, "negative" for a conjugate. The strike list is this test in standing form.

Both failures look like clarity from the inside; that is why the tests are written down.

The involution. Two involutions are available: the mode-swap ($X \leftrightarrow Y$, fixed set the balance) and the sign-flip (componentwise negation, fixed point O). **The mode-swap is structural; the sign-flip is chart.** Three independent grounds:

1. *Grammar.* "–x obtains" puts an obtainer's grammar on a contrast. A mode's opposite is its comparison-partner, not a mirror-thing that separately obtains.
2. *Chart.* The mode-swap survives redrawing; the sign on a mode does not.
3. *Motion.* In the swept rendering, a half-turn about the balance axis sends $X \rightarrow Y$, $Y \rightarrow X$: the mode-swap is a rotation, a motion the structure admits. Componentwise negation reverses orientation and is no rotation at all.

Everything resting on binding a component to its sign-image is therefore [rendering] (math chain). The signed register remains available as an indexed reading, which is what §3 says a sign is.

Open items

1. **The open closures** — the reciprocity joint's native-magnitudes step; anchoring-by-

constitution; the ground's self-sufficiency candidate.

2. **The cross-frame unit relation** $1_n \leftrightarrow 1_{(n+1)}$ — whether any quantitative relation accompanies the constitutive share of §7. Not needed for size comparison, which is not defined.
3. **Dimension count** — the orientation-freedom at the joint is a degree of freedom; whether it is space, and whether three, is not settled (two rival derivations in the math chain, unadjudicated).
4. **Branching beyond the balance.** Generation has no address (§6), so the question is whether any relation other than the constitutive pair can be inexact in the way that constitutes a child. Candidate, unruled: it cannot — the frame has one constitutive pair, and any conjugate equality within the frame reduces to the same balance. The step needing adversarial read is whether derivative pairings qualify.

v9.2 — the chains state the framework as currently known; the editorial record holds how it got here.